

Does Social Capital Strengthen the Stability of Household Income? Evidence from Cambodia

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Motivation

- ▶ Household income instability is typically defined as variations in **income across or within months/years** that are distinct from the overall income level.
 - Disrupt consumption smoothing (see, e.g., Günther and Harttgen 2009; Kazianga and Udry 2006; Ro et al. 1981)
 - Forcing households to adopt costly coping strategies such as distress asset sales (Dercon and Krishnan 2000), child labour withdrawal (Duryea et al. 2007), or high-interest debt (Collins et al. 2009).
 - These short-term responses often perpetuate long-term vulnerability, especially for low-income households.
 - Reinforces cycles of vulnerability and poverty traps (Dercon 2004; Carter and Barrett 2006).
 - Unstable income disproportionately burdens the poor who lack access to insurance, formal credits, or stable employment (Jalan and Ravallion 2004; Morduch 1995).

Motivation

- ▶ In that context, scholars and policymakers have turned to social capital as a potential informal mechanism to mitigate income instability.
 - Social capital is defined as “the features of social organisation, such as networks, norms, and trust that facilitate coordination and cooperation for mutual benefit” (Putnam et al. 1994, p.34), has been widely believed to mediate a household’s capacity to respond to adversity (Adger 2000; Pelling and High 2005; Aldrich and Meyer 2015; Zhang and Zhao 2024).
 - Social capital stabilises household income through two main channels:
 - First, **risk-sharing arrangements** (e.g., informal loans, reciprocal transfers, or kinship-based assistance) allow households to smooth consumption after income shocks (Townsend 1994; Fafchamps and Lund 2003).
 - Second, **social ties facilitate access to resources**, such as job referrals (Granovetter 1973), shared labour during planting seasons (Tuan et al. 2014), or collective savings groups (Anderson and Baland 2002).

Why does social capital matter?

- ▶ Despite its theoretical appeal, empirical evidence on the stabilising role of social capital remains unclear and context-dependent.
 - **Bonding social capital** (e.g., kinship ties) may help with small idiosyncratic shocks but can be less effective in systemic crises that affect entire communities (Portes 1998).
 - **Bridging or linking social capital**—connections to diverse groups or institutions—may offer stronger protection against covariate shocks, such as droughts or economic crises, by providing access to broader pools of resources (Woolcock and Narayan 2000; Grootaert et al. 2004).
 - Furthermore, **the effectiveness of social capital may depend on its distribution**: low-income households with weak networks may have limited ability to mobilise support, while high-income households' capture of social capital can reinforce inequality (Platteau 2015; Osborne et al. 2007).

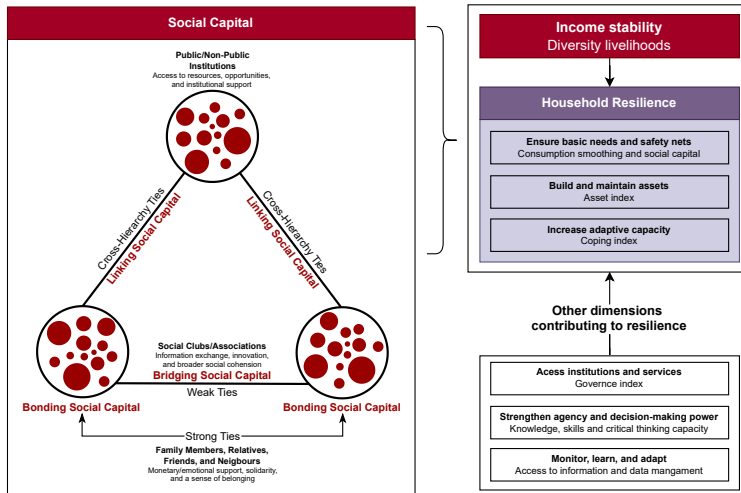
Research direction

- ▶ We aim to examine the role that social capital plays in strengthening the stability of household income and how these effects vary across different household characteristics in Cambodia.
 - **Data:** Using household panel data from rural villages collected by the Cambodia Development Resource Institute (CDRI) in four waves between 2011 and 2020.
 - **Focused on six demensions of social capital:** (1) number of trusted individuals; (2) number of association memberships; (3) percentage of trust in authorities; (4) percentage of participation in community; (5) percentage of households receiving social benefits; and (6) access to credit through social networks.
 - **Methodology:** OLS, instrument and quantile estimations.

Social capital in Cambodia

Period	Key Characteristics of Social Capital	Forms of Social Capital	Institutions/ Actors Involved	Trust and Reciprocity
Pre-1975 Pre-Khmer Rouge	• Strong kinship, religious, and village-based networks (e.g., pagodas) • Community cooperation through labour exchange and ceremonial support • Informal norms governed by elders and Buddhist values	Bonding	Monks, elders, kin networks, village leaders, pagodas	High interpersonal trust; moral and reciprocal obligations within communities
1975–1979 Khmer Rouge	• Systematic destruction of family and community bonds • Forced collectivisation and elimination of religious and traditional institutions • Breakdown of trust, widespread trauma (e.g., neighbours spied on each other)	Collapsed/ Destroyed	Khmer Rouge cadres, collectivised state apparatus	Extreme fear; erosion of trust; reciprocal norms dismantled
1980s–1990s Post-conflict	• Gradual rebuilding of trust and social ties • Emergence of local NGOs and village self-help groups • Return of religious practices and reformation of village institutions • Donor-led reconstruction	Bonding + Emerging Linking	Pagodas, local committees, NGOs, aid agencies	Rebuilding fragile trust; cautious reciprocity; external actors supplementing community ties
1990s Post-conflict recovery	• Proliferation of international NGOs and aid agencies • Participatory development and governance initiatives • Fragile cohesion reliant on external resources	Bonding + Linking	Commune councils, NGOs, donors, local development associations	Moderate trust in formal institutions; transactional reciprocity via projects
2000s Decentralisation and development	• Boost in local participation • Growth of civil society and community-based organisations • Decentralised governance connecting state and local actors	Bonding + Bridging + Linking	Commune councils, CSOs, village networks, donor programs	Improved trust through structured interaction; reciprocity linked to institutional roles
2010s Technology and urbanisation	• Rise of online and digital networks • Migration affects rural cohesion; new urban social structures emerge • Youth activism and digital mobilisation	Bridging + Hybrid (Digital–Informal)	Online communities, youth groups, NGOs, social media influencers	Peer trust increases; formal trust mixed; reciprocity via shared causes and digital interaction
2020s Resilience and uncertainty	• COVID-19 highlights mutual aid and community solidarity • Informal organising increases amid state shortfalls • The border conflict with Thailand increased mutual aid efforts for troops and evacuated people	Bonding + Digital Bridging/Linking	Mutual aid groups, informal leaders, grassroots organisations, diaspora networks	High informal trust; reliance on informal reciprocity; formal trust remains fragmented

Social capital and income instability framework



Survey sample

Table: Sample villages and attrition rates

Sample	2011	2020	Dropped out	Attrition rate
Andoung Trach	85	70	15	17.64%
Ba Boang	127	94	33	25.98%
Bo	85	82	3	3.52%
Dang Kdar	125	112	13	10.40%
Kanhchor	120	98	22	18.33%
Khsach Chi Ros	120	84	36	30.00%
Kompong Tnaot	120	107	13	10.83%
Krasang	120	90	30	25.00%
Prek Kmeng	120	105	15	12.50%
Prey Nobmuy	85	64	21	24.70%
Trapeang Prei	76	65	11	14.47%
All villages	1,183	971	212	17.92%

Descriptive statistics

Table: Summary Statistics: Baseline 2011 vs. Average 2011-2020

	2011					Average 2011-2020				
	Mean	SD	Min	Max	N	Mean	SD	Min	Max	N
Income instability	0.682	0.335	0.07	1.90	971	0.682	0.335	0.07	1.90	3884
Number of trusted people	2.021	1.066	0.00	4.00	971	2.065	1.091	0.00	4.00	3884
Number of associations	1.489	1.312	0.00	7.00	971	1.317	1.080	0.00	11.00	3884
Percent of trust in authorities	0.762	0.272	0.00	1.00	971	0.786	0.276	0.00	1.00	3884
Percent of social participation	0.206	0.134	0.00	0.60	971	0.145	0.129	0.00	0.75	3884
Percent of social beneficiary	0.235	0.125	0.05	0.65	971	0.191	0.125	0.00	0.80	3884
Access to SC finance	0.177	0.382	0.00	1.00	971	0.132	0.339	0.00	1.00	3884
Real household income (10,000 riel)	2198.478	3993.608	9.18	103792.58	971	3799.404	7870.636	0.00	193060.73	3884
ln(real household income)	7.243	0.943	2.22	11.55	971	7.559	1.151	2.22	12.17	3883
Real per capita income (10,000 riel)	437.581	551.146	9.18	11532.51	971	805.280	1662.086	0.00	48265.18	3884
Gender of head household	1.222	0.416	1.00	2.00	971	1.249	0.432	1.00	2.00	3884
Age of head household	49.986	12.493	19.00	88.00	971	53.529	12.689	16.00	95.00	3884
Year of education of head household	3.209	3.060	0.00	18.00	971	3.174	3.122	0.00	18.00	3884
Household size	4.967	1.951	1.00	15.00	971	4.859	2.040	1.00	15.00	3884
Dependency ratio	26.134	21.576	0.00	100.00	971	35.314	26.094	0.00	100.00	3884
Agricultural land size	2.074	2.891	0.00	40.40	971	2.308	3.415	0.00	53.50	3884

Empirical Results

	(1)	(2)	(3)	(4)	(5)	(6)
Number of trusted people	0.030*** (0.011)					
Number of associations		-0.035*** (0.008)				
Percent of trust in authorities			0.027 (0.039)			
Percent of social participation				-0.377*** (0.084)		
Percent of social beneficiary					-0.430*** (0.089)	
Access to SC finance						-0.012 (0.027)
ln(household income)	-0.041*** (0.014)	-0.037*** (0.014)	-0.034** (0.014)	-0.043*** (0.014)	-0.044*** (0.014)	-0.035*** (0.014)
Gender of head household	0.019 (0.028)	0.022 (0.029)	0.015 (0.029)	0.013 (0.028)	0.013 (0.028)	0.016 (0.029)
Age of head household	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000 (0.001)	0.000 (0.001)	0.001 (0.001)
Year of education of head household	0.009** (0.004)	0.009** (0.004)	0.009** (0.004)	0.008* (0.004)	0.008** (0.004)	0.009** (0.004)
Household size	-0.007 (0.006)	-0.004 (0.006)	-0.008 (0.006)	-0.002 (0.006)	-0.000 (0.006)	-0.007 (0.006)
Dependency ratio	0.001** (0.001)	0.001** (0.001)	0.001** (0.001)	0.001** (0.001)	0.001** (0.001)	0.001** (0.001)
Agricultural land size	-0.005 (0.003)	-0.001 (0.003)	-0.005 (0.003)	0.000 (0.003)	0.000 (0.003)	-0.005 (0.003)
Constant	0.838*** (0.109)	0.899*** (0.108)	0.834*** (0.114)	1.007*** (0.114)	1.027*** (0.113)	0.864*** (0.110)
R-squared	0.04	0.05	0.03	0.05	0.05	0.03
Observations	971	971	971	971	971	971

Note: Robust standard errors in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

OLS results

- ▶ The number of trusted people is positively and significantly associated with income instability.
- ▶ The number of associations a household belongs to is negatively and significantly related to income instability.
- ▶ The percentage of trust in authorities shows no significant correlation with income instability.
- ▶ The percentage of social participation and beneficiaries has a strong negative association with income instability.
- ▶ Access to credit through social capital does not appear to be associated with income instability.

	(1)	(2)	(3)	(4)	(5)	(6)
Number of trusted people	0.011 (0.030)					
Number of associations		-0.056*** (0.013)				
Percent of trust in local authorities			-0.026 (0.112)			
Percent of social participation				-0.622*** (0.119)		
Percent of social beneficiary					-0.705*** (0.130)	
Access to SC finance						-0.368** (0.153)
ln(household income)	-0.037** (0.015)	-0.038*** (0.014)	-0.035** (0.014)	-0.049*** (0.014)	-0.050*** (0.014)	-0.049*** (0.016)
Gender of head household	0.016 (0.028)	0.026 (0.029)	0.015 (0.028)	0.012 (0.028)	0.011 (0.028)	0.025 (0.031)
Age of head household	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
Year of education of head household	0.009** (0.004)	0.009** (0.004)	0.009** (0.004)	0.006 (0.004)	0.007* (0.004)	0.011** (0.004)
Household size	-0.007 (0.006)	-0.003 (0.006)	-0.007 (0.006)	0.002 (0.007)	0.004 (0.007)	-0.001 (0.007)
Dependency ratio	0.001** (0.001)	0.001* (0.001)	0.001** (0.001)	0.001** (0.001)	0.001** (0.001)	0.001 (0.001)
Agricultural land size	-0.005 (0.003)	0.002 (0.003)	-0.005 (0.003)	0.004 (0.004)	0.004 (0.004)	-0.005 (0.004)
Constant	0.851*** (0.109)	0.923*** (0.108)	0.883*** (0.155)	1.103*** (0.119)	1.134*** (0.120)	1.039*** (0.136)
Kleibergen-Paap LM p-val	0.000	0.000	0.000	0.000	0.000	0.000
Cragg-Donald F Stat	119.848	566.013	129.589	863.668	750.081	37.233
Observations	971	971	971	971	971	971

Note: Robust standard errors are reported in parentheses. *p<0.10, **p<0.05, ***p<0.01.

IV results

We constructed six instrumental variables based on the average level of social capital within the village, excluding the household's own social capital.

- ▶ The number of trusted people shows no statistically significant relationship with income instability.
- ▶ The results for the number of associations, participation, and benefits from social activities are consistent with the OLS models, differing only in magnitude and statistical significance.
- ▶ Access to social network credit appears to reduce income instability.

Table: Quantiles of income instability and social capital

	10th percentile	25th percentile	50th percentile	75th percentile	90th percentile
<i>Panel A</i>					
Number of trusted people	0.014 (0.013)	0.018* (0.010)	0.027** (0.011)	0.056*** (0.020)	0.052** (0.024)
<i>Panel B</i>					
Number of associations	-0.026*** (0.009)	-0.026*** (0.008)	-0.024*** (0.008)	-0.067*** (0.013)	-0.078*** (0.016)
<i>Panel C</i>					
Percent of trust in authorities	-0.023 (0.042)	0.050 (0.038)	0.029 (0.039)	0.038 (0.087)	0.009 (0.113)
<i>Panel D</i>					
Percent of social participation	-0.310*** (0.078)	-0.331*** (0.085)	-0.346*** (0.080)	-0.543*** (0.180)	-0.605*** (0.225)
<i>Panel E</i>					
Percent of social beneficiary	-0.360*** (0.082)	-0.394*** (0.096)	-0.371** (0.087)	-0.617*** (0.172)	-0.764*** (0.260)
<i>Panel F</i>					
Access to SC finance	-0.021 (0.033)	0.015 (0.028)	0.001 (0.023)	-0.067 (0.060)	-0.005 (0.071)

Note: All panels are based on 971 observations. Each specification controls for the natural logarithm of household income, household head's gender, age, years of education, household size, dependency ratio, and land size. Robust standard errors are reported in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Quantile regression estimates are based on 10,000 bootstrap repetitions.

Quantile regression results

- ▶ Trust helps buffer income shocks, especially for households with moderate-to-high volatility.
- ▶ Association memberships have consistently negative and significant effects across all quantiles, with stronger impacts (-0.067 to -0.078) at higher instability levels.
- ▶ Social participation and beneficiary status show the largest effects (-0.310 to -0.764), increasing at higher quantiles.
- ▶ Trust in authorities and access to social finance remain insignificant.

Discussion

- ▶ We find that four dimensions of social capital, namely, the number of association memberships, percentage of social participation, percentage of social beneficiaries, and access to social network finance, have a significant impact on household resilience, especially by reducing income instability.
- ▶ Close social ties (family, relatives, friends, neighbours) and institutional trust (in local and national authorities) show no statistically significant association with income instability. This suggests that while trust forms a necessary foundation for resilience, it does not directly generate income stability without being channelled into actionable support or institutional mechanisms.

Key takeaways

- ▶ Social capital contributes to reduce household income instability.
- ▶ While low-income households benefit from social capital, persistent poverty limits its stabilising effect. By contrast, high-income households are better able to leverage social capital to enhance income stability and economic security.
- ▶ From a policy perspective, the findings suggest:
 - The importance of strengthening both formal and informal social capital to support vulnerable households.
 - Investments in local associations, community participation mechanisms, and targeted benefit schemes can directly reduce income instability.
 - Fostering trust in public institutions through transparency, fairness, and rule-based governance can create an enabling environment in which social capital flourishes.

Key takeaways

- ▶ Additional reading: *Connecting communities to support Cambodia's vulnerable* (East Asia Forum, 13 August 2025). Available at:
<https://eastasiaforum.org/2025/08/13/connecting-communities-to-support-cambodias-vulnerable/>

Thank you!

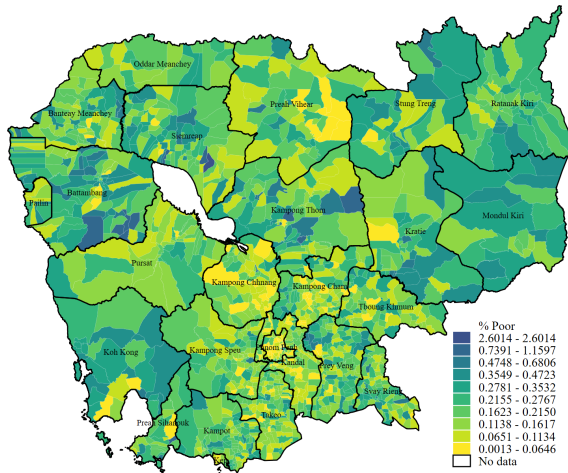
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Appendix

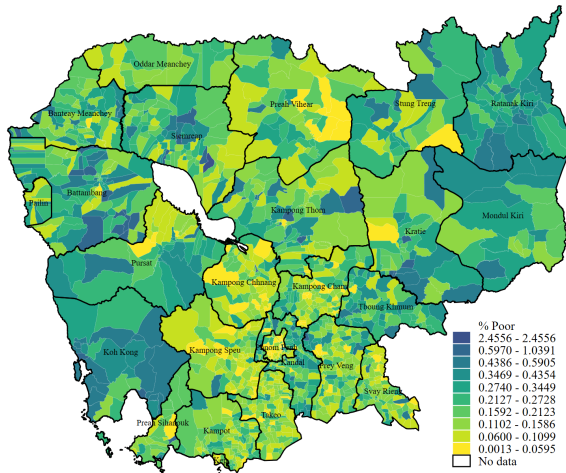
Figure: Proportion of poor households and individuals by commune

On average, 19.50 % of households in each commune were classified as poor.



Source: National Institute of Statistics of Cambodia

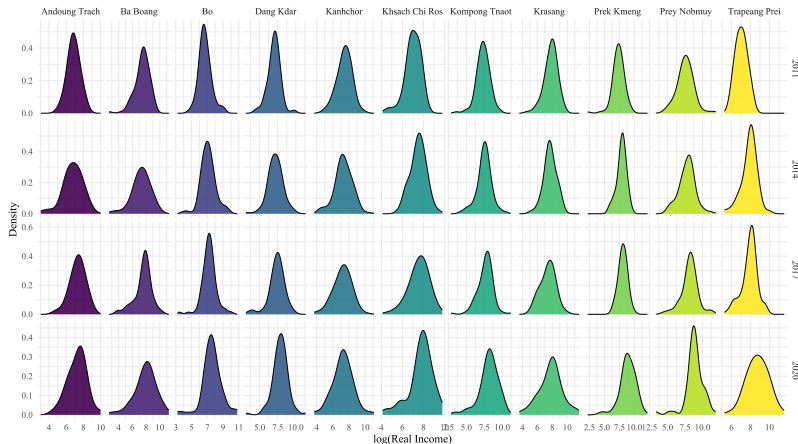
On average, 20.47 % of individuals in each commune were classified as poor.



Source: National Institute of Statistics of Cambodia

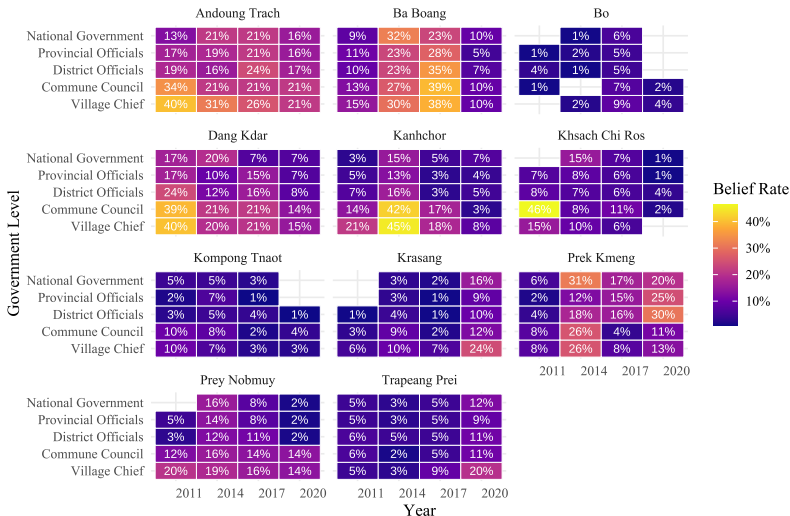
Note: The values displayed on these maps were in decimal form (ranging from 0 to 1), where 1 represents 100%.

Figure: Density of log(household real income) by year and village



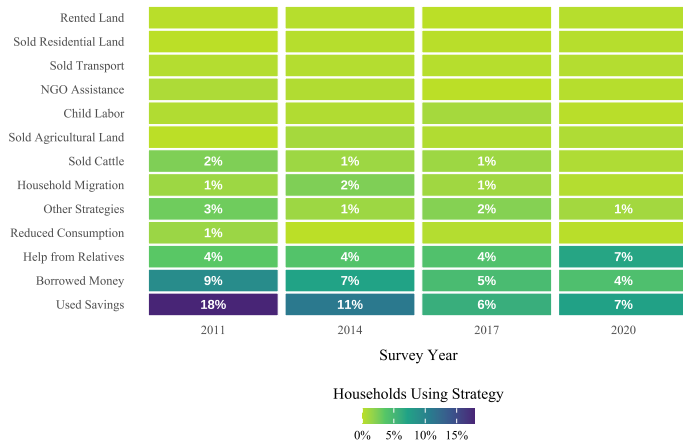
Note: The figure presents the distribution of log-transformed real household income by village and year, based on data from MOPS. Income is adjusted to constant 2008 prices using the national Consumer Price Index (CPI).

Figure: Population perception of trust in institutions



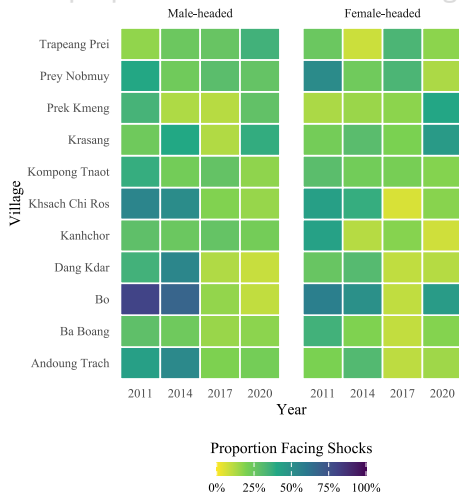
Note: Percentage shows households who answered "Believe" out of those who gave definite answers ("Believe" or "Not believe"). "Don't know" responses are excluded.

Figure: Household coping strategies



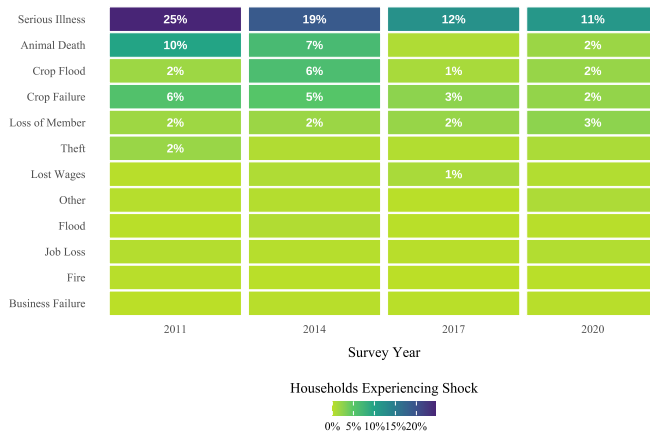
Note: This figure the prevalence of coping strategies adopted by households in response to shocks, disaggregated by year. The data are drawn from MOPS. Prevalence is calculated as the proportion of households reporting the use of each strategy in a given year.

Figure: The proportion of household facing shocks



The proportions are calculated using data from MOPS data. “Shock” refers to households reporting at least one adverse event, while “No Shock” indicates no such report.

Figure: Prevalence of household shocks



Prevalence is calculated as the mean proportion of households reporting each type of shock in a given year.